

Chapter 6

Electrostatic energy and Green's theorems

6.1 Electrostatic energy

Consider the work done by an external agent moving a charge q from point A to point B in an electrostatic field

$$W = -q \int_A^B \vec{E} \cdot d\vec{s} = q \int_A^B \vec{\nabla} \phi \cdot d\vec{s} = q (\phi(B) - \phi(A)) = q \phi(B) \quad (6.1)$$

if $\phi(A) = \phi(\infty) = 0$, as in the earlier examples of localized charge distributions.

The work needed to assemble a collection of charged particles is then

$$W = \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i}^N \frac{q_i q_j}{|\vec{x}_i - \vec{x}_j|} \quad (6.2)$$

For a continuous charge distribution, this becomes

$$W = \frac{1}{2} \iint \frac{\rho(\vec{x}') \rho(\vec{x})}{|\vec{x} - \vec{x}'|} dV dV' \quad (6.3)$$

Using $\rho = -\nabla^2\phi/4\pi$, we can rewrite this as

$$\begin{aligned}
 W &= \frac{1}{2} \int \phi(\vec{x}) \rho(\vec{x}) dV = -\frac{1}{8\pi} \int \phi \nabla^2 \phi dV \\
 &= -\frac{1}{8\pi} \int \left(\vec{\nabla} \cdot (\phi \vec{\nabla} \phi) - (\vec{\nabla} \phi)^2 \right) dV \\
 &= \frac{1}{8\pi} \int \vec{E}^2 dV
 \end{aligned} \tag{6.4}$$

dropping the total divergence, whose surface term vanishes at infinity. We conclude that

$$u = \frac{E^2}{8\pi} = \text{electrostatic energy density} \tag{6.5}$$

6.2 Statics and conductors

Inside a conductor in electrostatics

$$\vec{E} = 0 \quad , \quad \phi = \text{constant} \tag{6.6}$$

Given N conductors with charges Q_i and potentials ϕ_i , we have the linear relation

$$Q_i = C_{ij} \phi_j \tag{6.7}$$

where the C_{ij} with $i \neq j$ are called capacitance coefficients, and the C_{ii} self-inductance coefficients.

As an example, consider two concentric spherical shells of radii a and $b > a$, carrying charges Q_1 and Q_2 respectively, as shown in the Figure. The

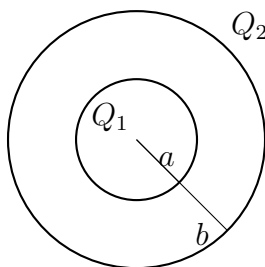


Figure 6.1: Two concentric spherical shells of radii a and b , carrying charges Q_1 and Q_2 .

potentials on the two shells are

$$\phi_b = \frac{Q_1}{b} + \frac{Q_2}{b} \quad , \quad \phi_a = \frac{Q_1}{a} + \frac{Q_2}{b} \quad (6.8)$$

which we can invert

$$\begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = \frac{b}{a-b} \begin{pmatrix} a & -a \\ -a & a \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} \quad (6.9)$$

Setting $Q_1 = Q$, $Q_2 = -Q$ defines the capacitance

$$C \equiv \frac{Q}{|\phi_1 - \phi_2|} > 0 \quad , \quad C = \frac{ab}{b-a} \quad (6.10)$$

6.3 Green's theorems

We have seen that

$$\phi(\vec{x}) = \int \frac{\rho(\vec{x}') dV'}{|\vec{x} - \vec{x}'|} \quad (6.11)$$

solves $\nabla^2 \phi = -4\pi\rho$ as demonstrated, but with the boundary condition $\phi \rightarrow 0$ at infinity. We want a more general solution: a region where $\nabla^2 \phi = -4\pi\rho$ holds and ϕ is unknown, with the outside unknown as well, but where we know ϕ or $\partial\phi/\partial n = E_n$ at the boundary (n being the normal direction to the boundary), as depicted in the Figure. To get there, consider Gauss's law

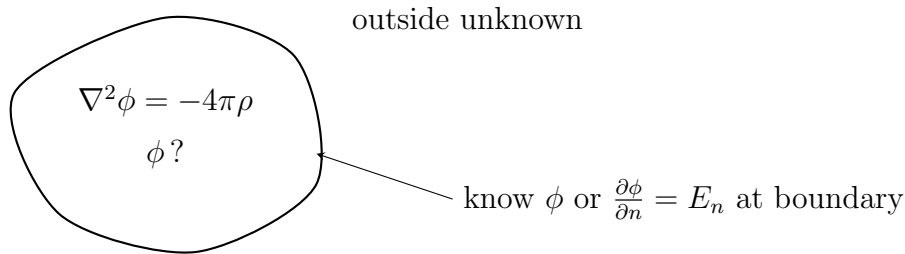


Figure 6.2: A region in which $\nabla^2 \phi = -4\pi\rho$, with ϕ or $\partial\phi/\partial n$ known at the boundary and everything outside unknown.

$$\int \vec{\nabla} \cdot \vec{G} dV = \oint \vec{G} \cdot d\vec{a} \quad (6.12)$$

and let $\vec{G} = \varphi \vec{\nabla} \psi$, so that

$$\vec{\nabla} \cdot \vec{G} = \varphi \nabla^2 \psi + \vec{\nabla} \varphi \cdot \vec{\nabla} \psi \quad (6.13)$$

This gives

$$\int \left(\varphi \nabla^2 \psi + \vec{\nabla} \varphi \cdot \vec{\nabla} \psi \right) dV = \oint \varphi \vec{\nabla} \psi \cdot d\vec{a} = \oint \varphi \frac{\partial \psi}{\partial n} da \quad (6.14)$$

Exchanging $\varphi \leftrightarrow \psi$ and subtracting, we get Green's theorem

$$\int \left(\varphi \nabla^2 \psi - \psi \nabla^2 \varphi \right) dV = \oint \left(\varphi \vec{\nabla} \psi - \psi \vec{\nabla} \varphi \right) \cdot d\vec{a} \quad (6.15)$$

The theorem can be used to prove existence and uniqueness of the solution to $\nabla^2 \phi = -4\pi\rho$ with given boundary conditions.

First, consider

$$\nabla^2 G(\vec{x}, \vec{x}') = -4\pi \delta^3(\vec{x} - \vec{x}') \quad (6.16)$$

with either

$$G = 0 \quad \text{at the boundary (Dirichlet)} \rightarrow G \equiv G_D \quad (6.17)$$

or

$$\frac{\partial G}{\partial n} = -\frac{4\pi}{S} \quad \text{at the boundary (Neumann)} \rightarrow G \equiv G_N \quad (6.18)$$

with S the total area of the boundary. (Why $\partial G/\partial n = -4\pi/S$? We might want $\partial G/\partial n = 0$, but this is not possible:

$$\int \nabla^2 G dV = \int \vec{\nabla} \cdot \vec{\nabla} G dV = -4\pi \Rightarrow \oint \vec{\nabla} G \cdot d\vec{a} = \oint \frac{\partial G}{\partial n} da = -4\pi \quad (6.19)$$

which rules out a vanishing normal derivative everywhere on the boundary.)

We want the solution to this problem, that is we need to find $G(\vec{x}, \vec{x}')$. Write

$$G(\vec{x}, \vec{x}') = \frac{1}{|\vec{x} - \vec{x}'|} + F(\vec{x}, \vec{x}') \quad (6.20)$$

– the Green's function – where

$$\nabla^2 F = 0 \quad (6.21)$$

We need F so that the appropriate boundary conditions are satisfied. Note that from (6.16) we expect

$$G(\vec{x}, \vec{x}') = G(\vec{x}', \vec{x}) \quad , \quad G(\vec{x}, \vec{x}') = G(|\vec{x} - \vec{x}'|) \quad (6.22)$$

because of the rotational and translational symmetry of the equation.

Given $G(\vec{x}, \vec{x}')$, we get ϕ as follows. Using Green's theorem (6.15), set $\varphi = \phi$ and $\psi = G(\vec{x}, \vec{x}')$, with the integration taken over the primed variable

$$\begin{aligned} & \int [\phi(\vec{x}') 4\pi \delta^3(\vec{x} - \vec{x}') + G(\vec{x}, \vec{x}') (-4\pi \rho(\vec{x}'))] dV' \\ &= \oint \left(G(\vec{x}, \vec{x}') \frac{\partial \phi(\vec{x}')}{\partial n'} - \phi(\vec{x}') \frac{\partial G(\vec{x}, \vec{x}')}{\partial n'} \right) da' \end{aligned} \quad (6.23)$$

which leads to

$$\begin{aligned} \phi(\vec{x}) &= \int G(\vec{x}, \vec{x}') \rho(\vec{x}') dV' \\ &+ \frac{1}{4\pi} \oint \left(G(\vec{x}, \vec{x}') \frac{\partial \phi(\vec{x}')}{\partial n'} - \phi(\vec{x}') \frac{\partial G(\vec{x}, \vec{x}')}{\partial n'} \right) da' \end{aligned} \quad (6.24)$$

In the Dirichlet case, the first surface term vanishes since $G_D = 0$ at the boundary, and we need to know ϕ at the boundary. In the Neumann case, $\partial G / \partial n' = -4\pi / S$ turns the second surface term into

$$\frac{1}{4\pi} \frac{4\pi}{S} \oint \phi da' = \bar{\phi} = \text{constant} \quad (6.25)$$

the average of ϕ over the boundary, and we need to know $\partial \phi / \partial n$ at the boundary. All in all,

$$G = (\text{part for charges inside}) + (\text{part for charges outside, through b.c.}) \quad (6.26)$$

where the first part is the $1/|\vec{x} - \vec{x}'|$ piece, and the second is F .

6.3.1 Two dimensions

In the two-dimensional case, we have

$$\nabla^2 G(\vec{x}, \vec{x}') = -4\pi \delta^2(\vec{x} - \vec{x}') \quad (6.27)$$

with Dirichlet or Neumann boundary conditions at the boundary. Once again

$$G(\vec{x}, \vec{x}') = G_{\text{inside}}(\vec{x}, \vec{x}') + F(\vec{x}, \vec{x}') \quad (6.28)$$

and Green's theorem then gives us ϕ . Symmetry implies $G(\vec{x}', \vec{x}) = G(r)$ where $\vec{r} = \vec{x} - \vec{x}'$ in polar coordinates, so that

$$\nabla^2 G = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) G(r) = \delta^2(\vec{x}) = \frac{\delta(r)}{2\pi r} \quad (6.29)$$

since $\int \delta^2(\vec{x}) d^2x = 1$, which gives

$$G = \frac{1}{2\pi} \ln r \quad (6.30)$$

Hence

$$G(\vec{x}, \vec{x}') = \frac{1}{2\pi} \ln |\vec{x} - \vec{x}'| + F(\vec{x}, \vec{x}') \quad (6.31)$$

and

$$\begin{aligned} \phi(\vec{x}) &= \int G(\vec{x}, \vec{x}') \rho(\vec{x}') da' \\ &+ \frac{1}{4\pi} \oint \left(G(\vec{x}, \vec{x}') \frac{\partial \phi(\vec{x}')}{\partial n'} - \phi(\vec{x}') \frac{\partial G(\vec{x}, \vec{x}')}{\partial n'} \right) ds' \end{aligned} \quad (6.32)$$

where the last term is a line integral along the boundary.

6.3.2 One dimension

In one dimension

$$\nabla^2 G = -4\pi \delta(x - x') \quad (6.33)$$

which gives

$$G(x, x') = \frac{1}{2} (x - x') [\Theta(x - x') - \Theta(x' - x)] + F(x, x') \quad (6.34)$$

using

$$\frac{d\Theta(x)}{dx} = \delta(x) \quad (6.35)$$

The boundary is now two points.